FEMI SAMUEL ADEOLA

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Research Profile Summary

Applied AI researcher interested in AI governance, trustworthy AI, and human-centred decision-support systems. My work examines how AI systems can be designed, evaluated, and governed for responsible use in public-sector and institutional contexts—combining hands-on experience building and evaluating AI systems with a developing focus on the governance, accountability, and organisational dimensions of their deployment. I am particularly interested in the conditions under which institutions can responsibly trust AI-generated outputs, and how governance can be built into system design from the outset rather than applied after the fact. I take a design-oriented research approach to building and evaluating practical AI solutions to real-world problems.

Research Interests

- **AI Governance & Accountability:** Institutional trust, regulatory compliance, and responsible deployment of AI in public sector and high-stakes environments.
- **Human-Centred AI & Decision-Support:** Human-in-the-loop systems, user trust, and the organisational dimensions of AI adoption.
- **Trustworthy & Explainable AI (XAI):** Transparency, fairness, and auditability in AI-assisted decision-making; interpretability–accuracy trade-offs.
- **AI in Education & Public Sector:** Educational data mining, programme evaluation, and policy-relevant AI applications.

Education

M.Sc. in Computer Engineering **Oct 2024 – Dec 2026 (expected)**
Federal University of Rio Grande (FURG), Brazil

- **Focus:** Foundations and Applications of Artificial & Computational Intelligence.
- **Dissertation:** *Grounding LLMs in Institutional Data: A Neuro-Symbolic RAG Framework for Postgraduate Program Evaluation*
- **Research Scope:** Built and evaluated an AI-assisted decision-support system for educational policy, integrating GPT-4o, Gemini 2.5 Flash, and DeepSeek with LangChain and FAISS to evaluate Brazilian postgraduate programmes across three CAPES quadrennial cycles (2013–2024). Retrieval-augmented settings showed substantially better groundedness than ungrounded baselines.

B.Sc. in Computer Science **2013 – 2019**
National Open University of Nigeria (NOUN), Nigeria

- **Project:** Design and Implementation of a Secondary School Management Information System (Case Study: Decency Education Centre, Benin City, Nigeria).
- **Key Achievement:** Built and deployed a centralised administrative platform to manage student data for a local educational institution.

Research Experience

Graduate Researcher (Applied AI) **Oct 2024 – Present**
Federal University of Rio Grande (FURG)

- **Decision-Support System Design:** Built and evaluated an AI-assisted decision-support system for educational policymakers, using RAG to ground institutional analysis in verified data across three quadrennial evaluation cycles.

- **Evaluation Across Institutional Contexts:** Assessed system outputs across 220 programmes spanning diverse regional and institutional profiles, analysing prediction reliability by grade tier and programme type — surfacing where AI-generated evaluations aligned or diverged from official committee decisions.
- **Stakeholder-Oriented Evaluation:** Built evaluation frameworks measuring output reliability, fairness across institutional demographics, and transparency of reasoning for non-technical users — combining groundedness, adjacent accuracy (± 1 grade tolerance), and symbolic interpretability metrics.
- **Technical Implementation:** Built the RAG pipeline (LangChain, FAISS, GPT-4o, Gemini 2.5 Flash, DeepSeek) underpinning the evaluation framework.

Teaching Internship

Teaching Internship in Undergraduate Studies **June 2026**
Federal University of Rio Grande (FURG) — supervised by Prof. Dr. Eduardo Nunes Borges

- Taught retrieval-augmented generation as an extension of classical Information Retrieval, covering dense embeddings, FAISS indexing, and LLM grounding to a Computing cohort versed in vector space models and cosine similarity.
- Supported a structured class assessment requiring critical evaluation and proposed improvements to a live RAG system.

Journal and Conference Manuscripts

- **Adeola, F. S.**, & Saxena, S. (2026). *A Retrospective Metadata Analysis of Data.gov: US' Open Government Data (OGD) Portal. Quality & Quantity* (Springer Nature). **[Accepted]**
- **Adeola, F. S.**, & Saxena, S. (2026). *Investigating the Progress of Open Government Data (OGD) Publishing in Brazil.* Submitted to *Scientometrics*. **[Under Review]**
- **Adeola, F. S.**, et al. (2026). *Balancing Accuracy and Interpretability: A Neuro-Symbolic RAG Framework for Educational Policy Evaluation.* Submitted to the 19th International Conference on Computer Supported Education (CSEDU 2027). **[Submitted]**
- **Adeola, F. S.**, et al. (2026). *Where Symbolic Rules End and LLMs Begin: A Hybrid Neuro-Symbolic RAG Framework for Programme Prediction.* Manuscript prepared for journal submission. **[In Preparation]**

Selected Research Projects

AI Decision-Support Framework for Policy Evaluation
Designed and evaluated a governance-aligned AI decision-support system for educational policymakers, combining a deterministic rule engine with RAG-based neural scoring to predict and explain CAPES quadrennial evaluation grades for 220 Brazilian postgraduate programmes. Live prototype deployed on HuggingFace Spaces.

AI in Finance: Trustworthy RAG for Claims Processing
Designed a retrieval-augmented generation system linking policy documents with claim inquiries, focused on accurate, legally compliant retrieval in a high-stakes financial context.

Open Government Data Analysis
Case study of OGD publishing trends in the USA, connecting data transparency practice to AI accountability themes.

Technical Skills

- **AI & RAG Frameworks:** LangChain, LlamaIndex, FAISS, embedding generation.
- **Data Science:** Python (Pandas, NumPy), SQL, Power BI.
- **Tools & Cloud:** Jupyter Notebooks, Git/GitHub, Google Cloud Platform (GCP).

Professional Experience

Web & Data Systems Specialist **Mar 2020 – Oct 2024**
Carelifeline Limited, UK (Remote)

- Managed digital information systems and contributed data-informed insights to organisational decision-making, including recruitment and operational reporting.
- Built experience in responsible data handling, system documentation, and communicating analytical outputs to non-technical stakeholders.

Classroom Teacher & Curriculum Support **Sep 2010 – Dec 2013**
New Discovery High School, Nigeria

- Designed and delivered structured lesson plans, integrating technology to support learning and engagement.
- Adapted instructional approaches to meet the diverse learning needs of students in basic education.

Languages

- **English:** Native | **Portuguese:** Basic

Certifications & Memberships

- **Certifications:** Responsible Generative AI (Uni. of Michigan); Generative AI Engineering (IBM).
- **Memberships:** Association for Computing Machinery — ACM SIGAI; Association for the Advancement of AI (AAAI). *(AIS student membership)*

References

- **Prof. Dr. Eduardo Nunes Borges** — Supervisor, Center for Computational Sciences (C3), Federal University of Rio Grande (FURG), Brazil.
- **Prof. Dr. Rodrigo Andrade de Bem** — Co-Supervisor, Center for Computational Sciences (C3), Federal University of Rio Grande (FURG), Brazil.
- **Prof. Stuti Saxena** — Co-author, Dept. of ICEAC, Federal University of Rio Grande (FURG), Brazil.